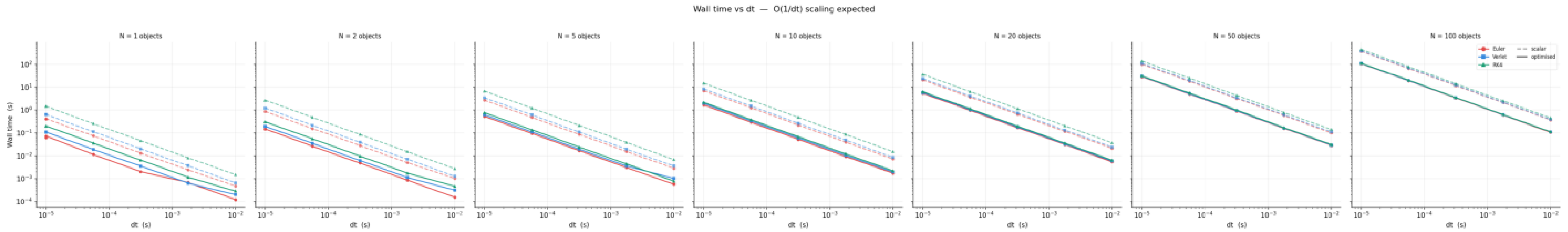
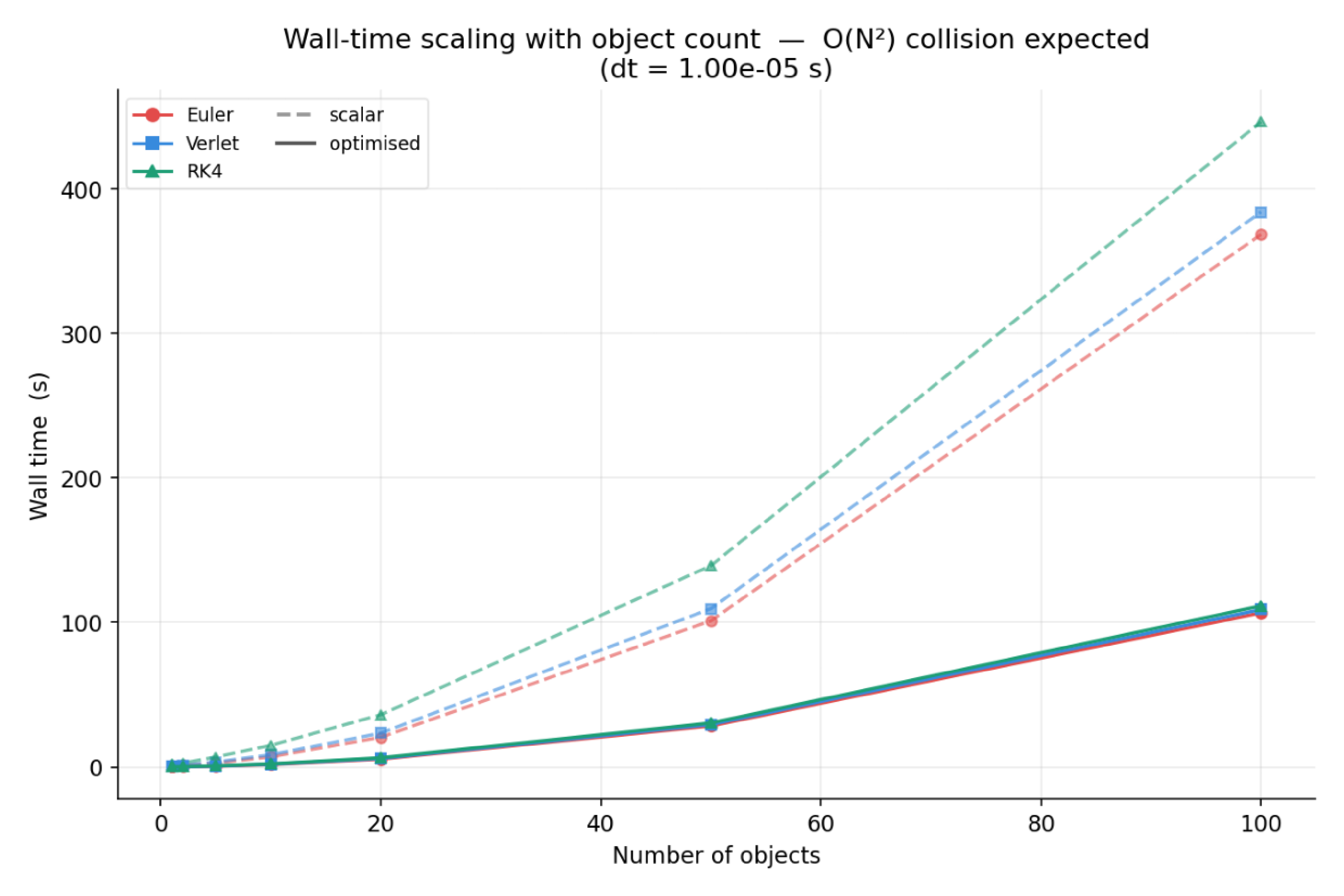
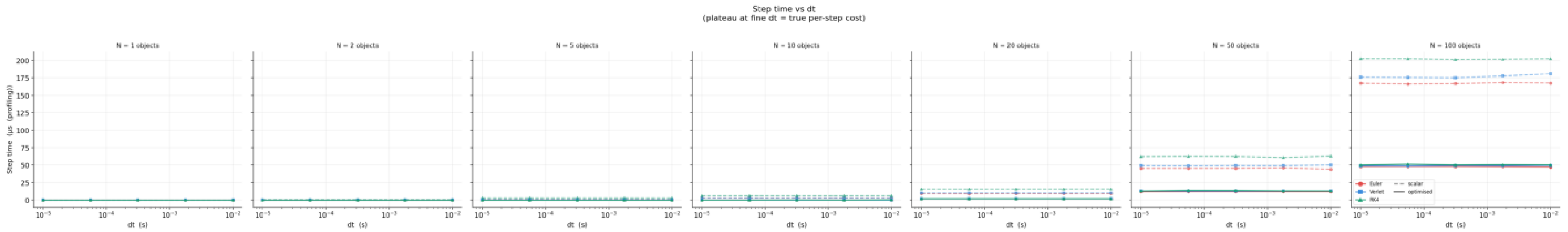


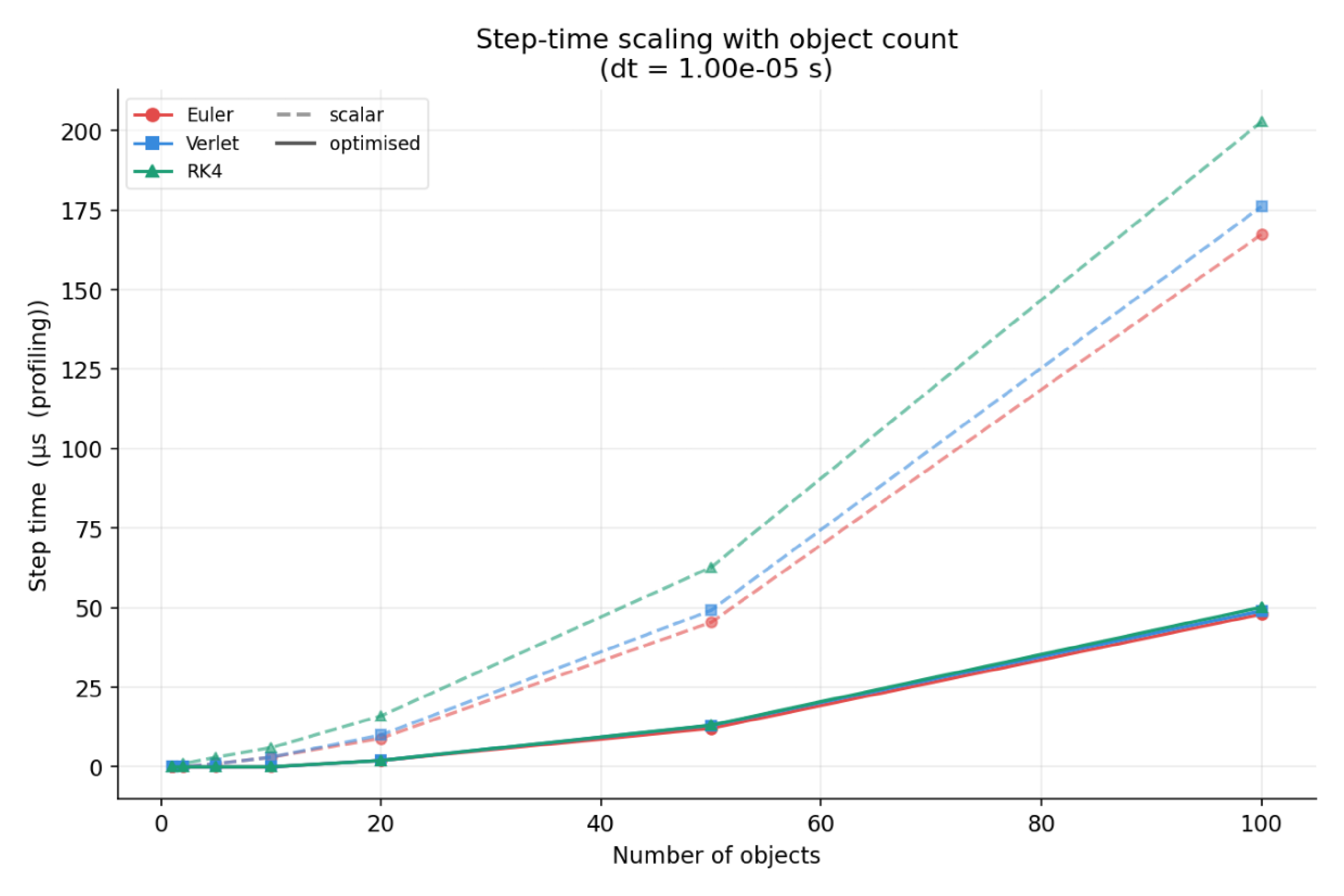
Wall time vs dt — O(1/dt) scaling expected
N = 1 objects
N = 2 objects
N = 5 objects
N = 10 objects
N = 20 objects
N = 50 objects
N = 100 objects
Wall time (s)
dt (s)
Euler
Verlet
RK4
scalar
optimised



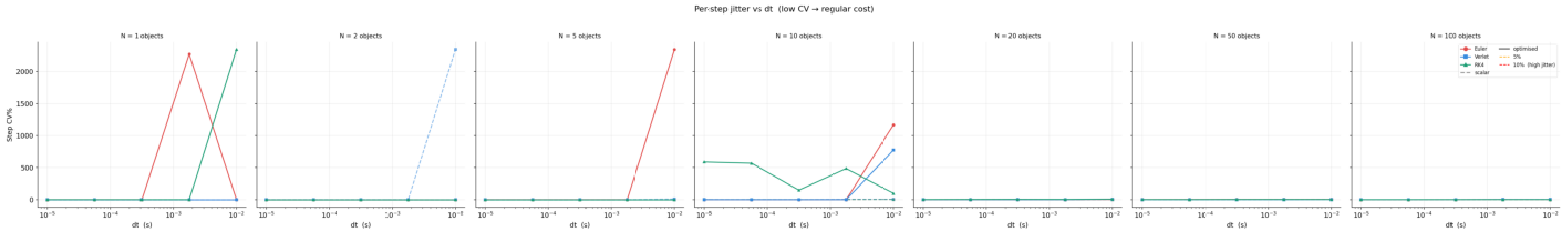
Wall-time scaling with object count — O(N²) collision expected
(dt = 1.00e-05 s)
Euler
Verlet
RK4
scalar
optimised
Wall time (s)
Number of objects
0
100
200
300
400
0
20
40
60
80
100



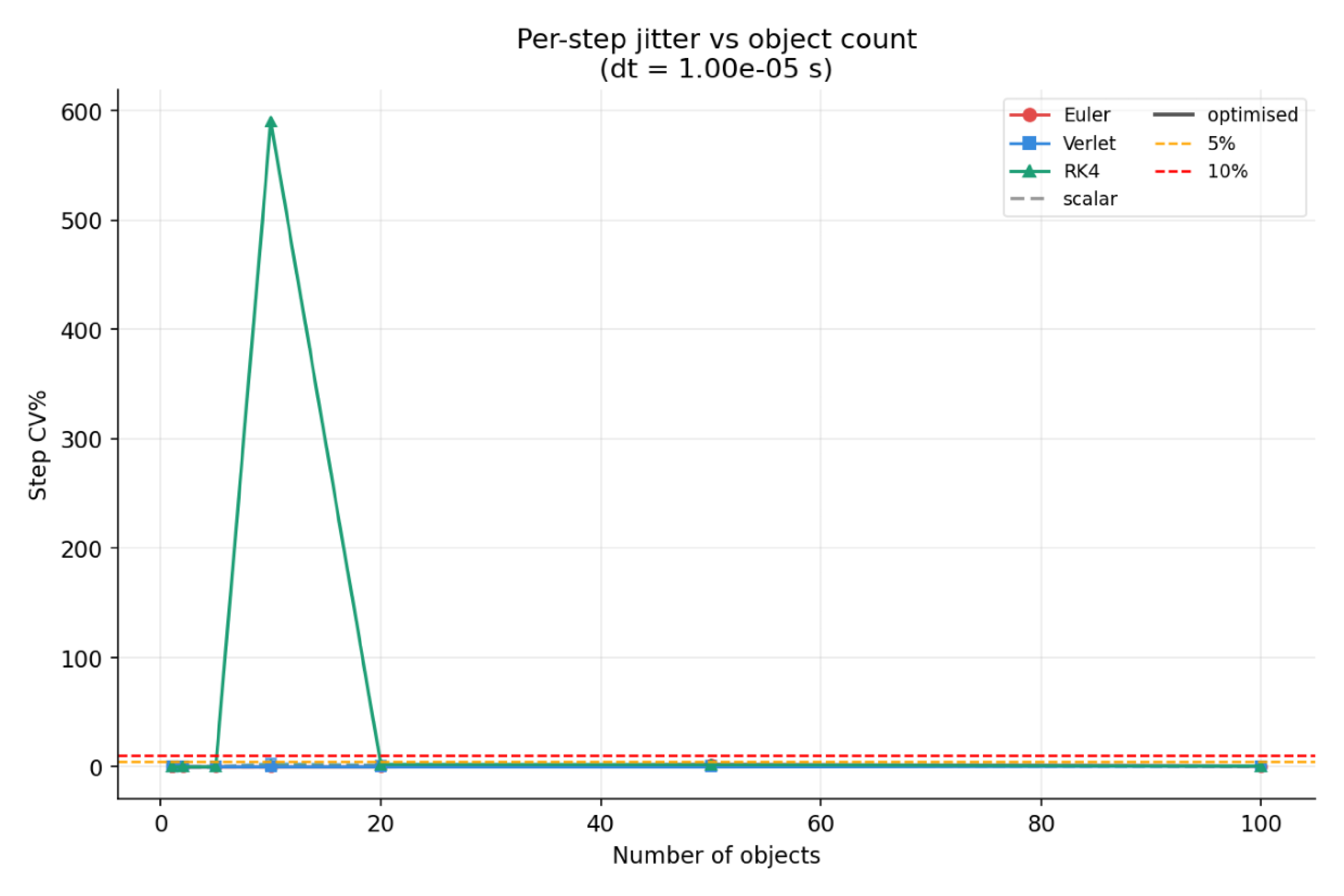
Step time vs dt
(plateau at fine dt = true per-step cost)
N = 1 objects
N = 2 objects
N = 5 objects
N = 10 objects
N = 20 objects
N = 50 objects
N = 100 objects
Step time (µs (profiling))
dt (s)
Euler
Verlet
RK4
scalar
optimised

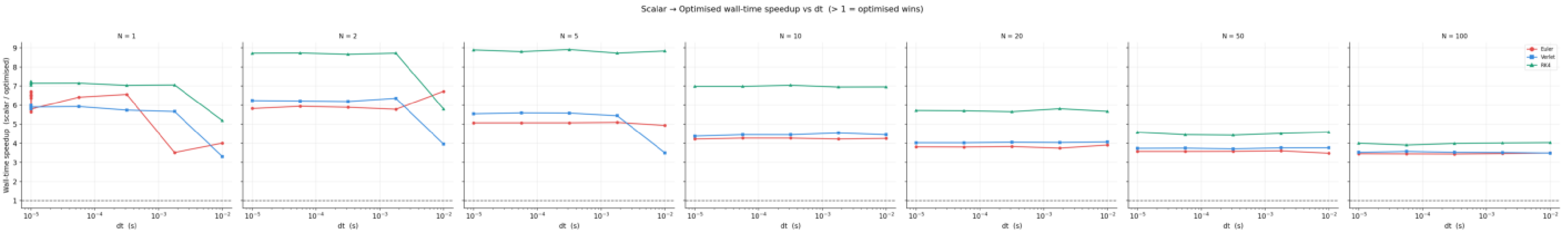


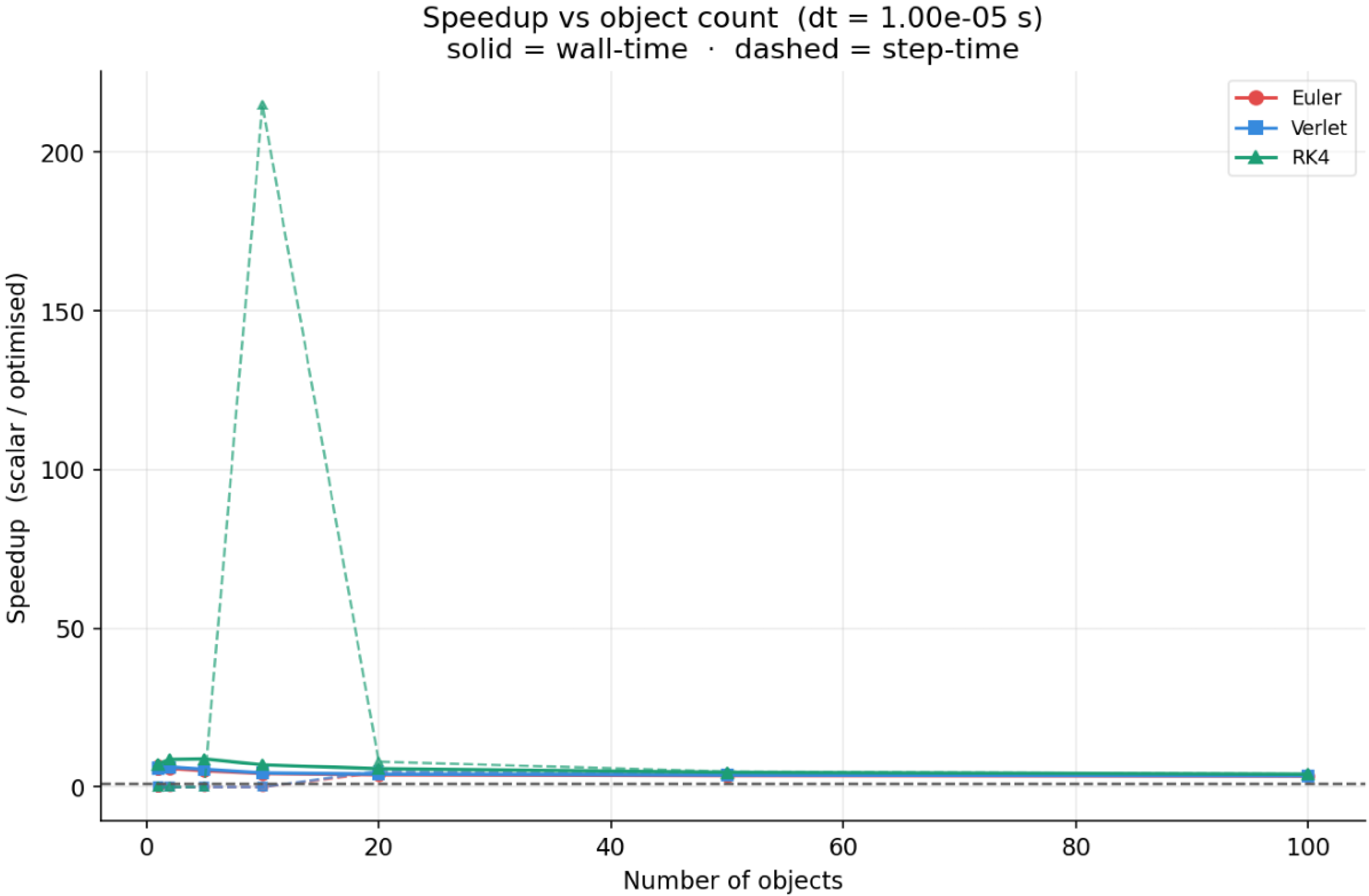
Step-time scaling with object count
(dt = 1.00e-05 s)
Euler
Verlet
RK4
scalar
optimised
Step time (µs (profiling))
Number of objects
0
25
50
75
100
125
150
175
200
0
20
40
60
80
100

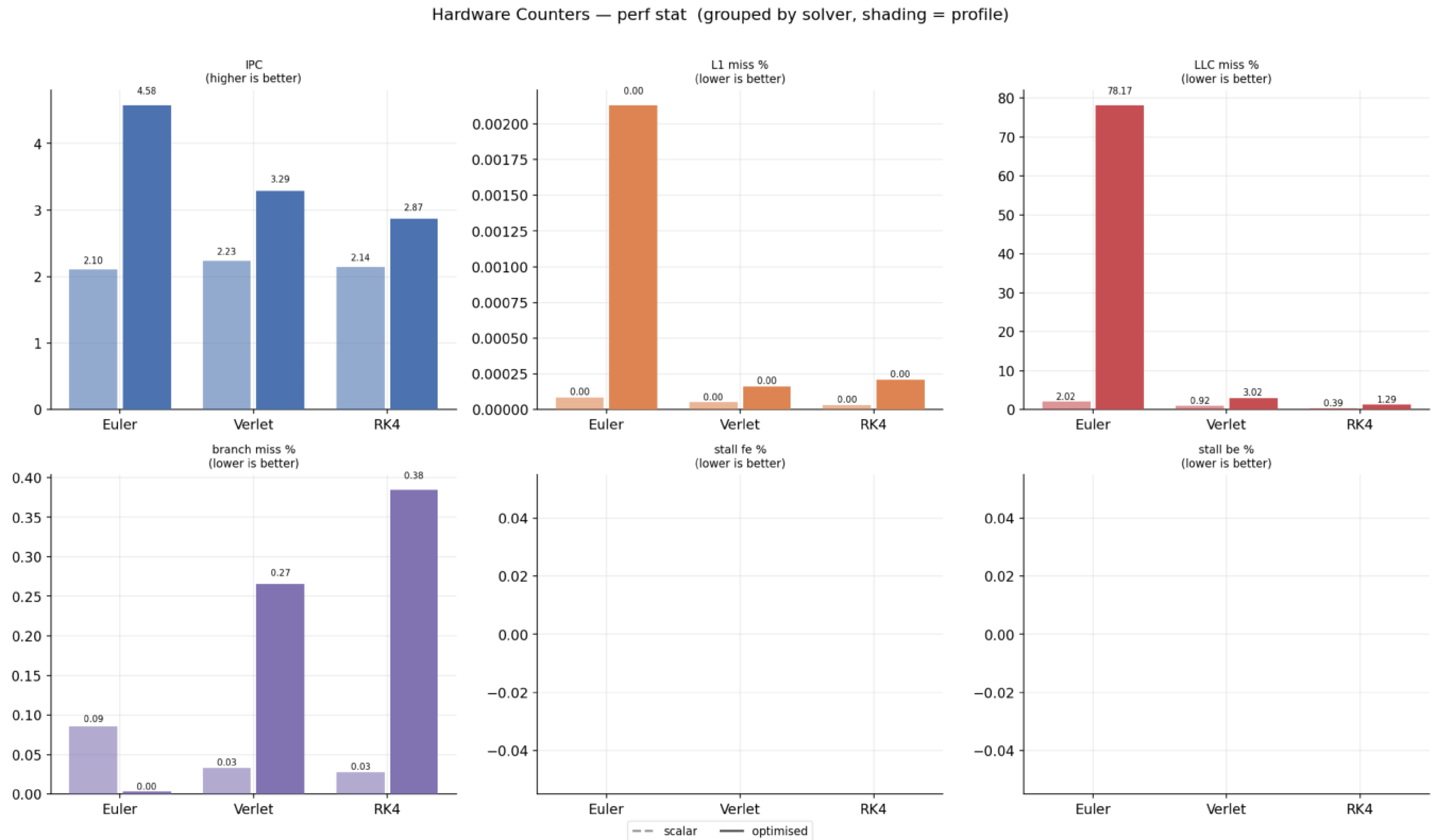


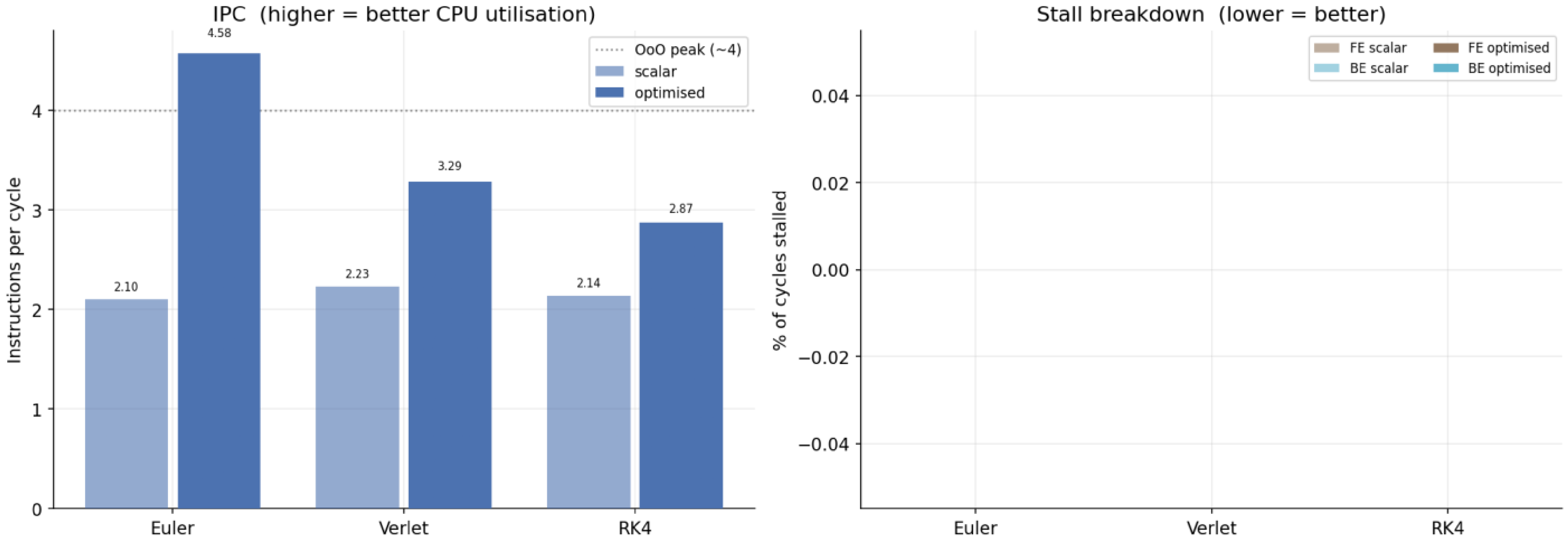
Per-step jitter vs dt (low CV → regular cost)
N = 1 objects
N = 2 objects
N = 5 objects
N = 10 objects
N = 20 objects
N = 50 objects
N = 100 objects
Step CV%
0
500
1000
1500
2000
dt (s)
Euler
Verlet
RK4
scalar
optimised
5%
10% (high jitter)


Per-step jitter vs object count
(dt = 1.00e-05 s)
Step CV%
Number of objects
0
100
200
300
400
500
600
0
20
40
60
80
100
Euler
Verlet
RK4
scalar
optimised
5%
10%

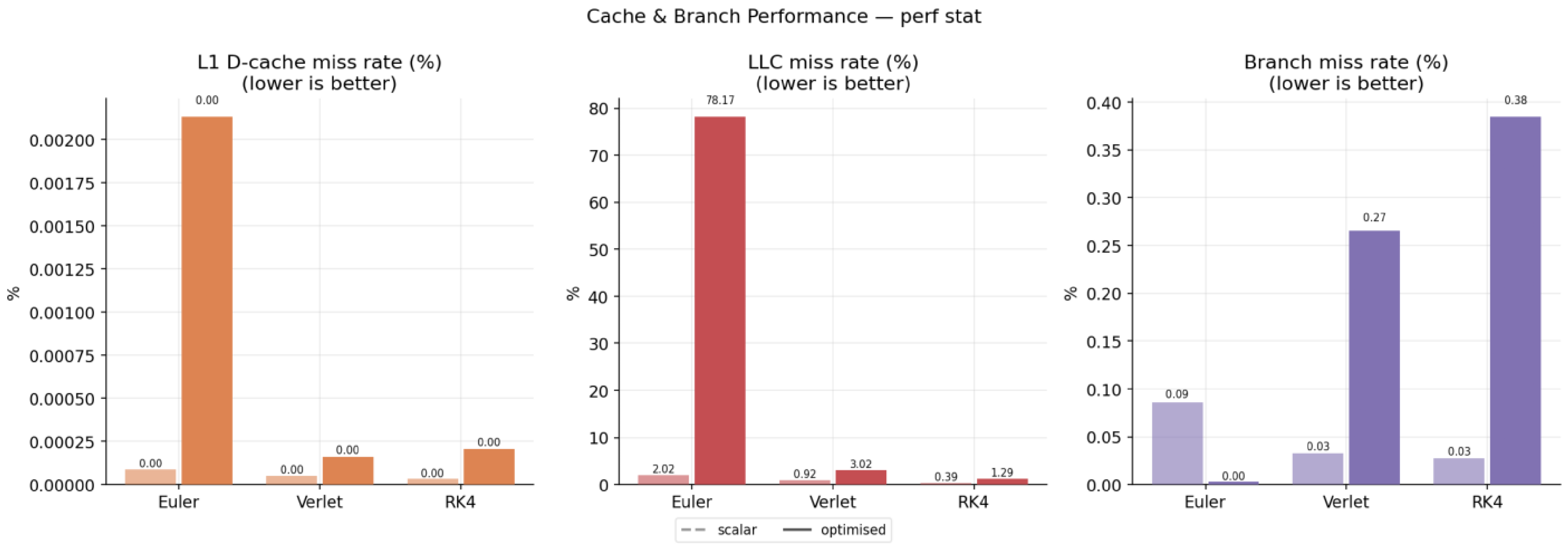

Scalar → Optimised wall-time speedup vs dt (> 1 = optimised wins)
N = 1
N = 2
N = 5
N = 10
N = 20
N = 50
N = 100
Wall-time speedup (scalar / optimised)
dt (s)
Euler
Verlet
RK4


Speedup vs object count (dt = 1.00e-05 s)
solid = wall-time · dashed = step-time
Euler
Verlet
RK4
200
150
100
50
0
0
20
40
60
80
100
Speedup (scalar / optimised)
Number of objects


Hardware Counters — perf stat (grouped by solver, shading = profile)
IPC
(higher is better)
4.58
3.29
2.87
2.10
2.23
2.14
L1 miss %
(lower is better)
0.00
LLC miss %
(lower is better)
78.17
2.02
0.92
3.02
0.39
1.29
branch miss %
(lower is better)
0.38
0.27
0.09
0.00
0.03
0.03
stall fe %
(lower is better)
stall be %
(lower is better)
Euler
Verlet
RK4
scalar
optimised



IPC (higher = better CPU utilisation)
Instructions per cycle
OoO peak (~4)
scalar
optimised
4.58
2.10
2.23
3.29
2.14
2.87
0
1
2
3
4
Euler
Verlet
RK4
Stall breakdown (lower = better)
% of cycles stalled
FE scalar
FE optimised
BE scalar
BE optimised
0.04
0.02
0.00
−0.02
−0.04
Euler
Verlet
RK4


Cache & Branch Performance — perf stat
L1 D-cache miss rate (%)
(lower is better)
LLC miss rate (%)
(lower is better)
Branch miss rate (%)
(lower is better)
%
Euler
Verlet
RK4
0.00
2.02
78.17
0.92
3.02
0.39
1.29
0.09
0.03
0.27
0.38
scalar
optimised